

Clinical Digital Twin Report

Computational Cellular Reprogramming Analysis

Session ID: session_2026-01-06T04-19-43-964Z
Run ID: #4
Date Generated: 2026-01-06 04:19:51
Total Cells: 1987 | Model Mode: CLINICAL | Avg BioAge: N/A

1. Introduction

Cellular reprogramming and transdifferentiation offer promising avenues for regenerative medicine by converting somatic cells into other cell types. The objective of this experiment, Run ID: #4, was to evaluate the efficacy of direct cardiac transdifferentiation and OSKM-mediated (Yamanaka factors) reprogramming in altering the cellular fate of somatic populations.

2. Materials and Methods

The experiment employed a combination of computational analysis using scVI and a series of cellular interventions, including direct cardiac transdifferentiation and multiple OSKM transfections, interspersed with oncogenic stress induction. These methodologies were applied systematically to assess their impact on cellular reprogramming.

Experimental Interventions:

- Direct Cardiac Transdifferentiation
- OSKM Transfection (Yamanaka)
- Oncogenic Stress Induced
- Oncogenic Stress Induced
- OSKM Transfection (Yamanaka)
- OSKM Transfection (Yamanaka)
- OSKM Transfection (Yamanaka)
- OSKM Transfection (Yamanaka)
- Direct Cardiac Transdifferentiation
- Direct Cardiac Transdifferentiation
- Oncogenic Stress Induced

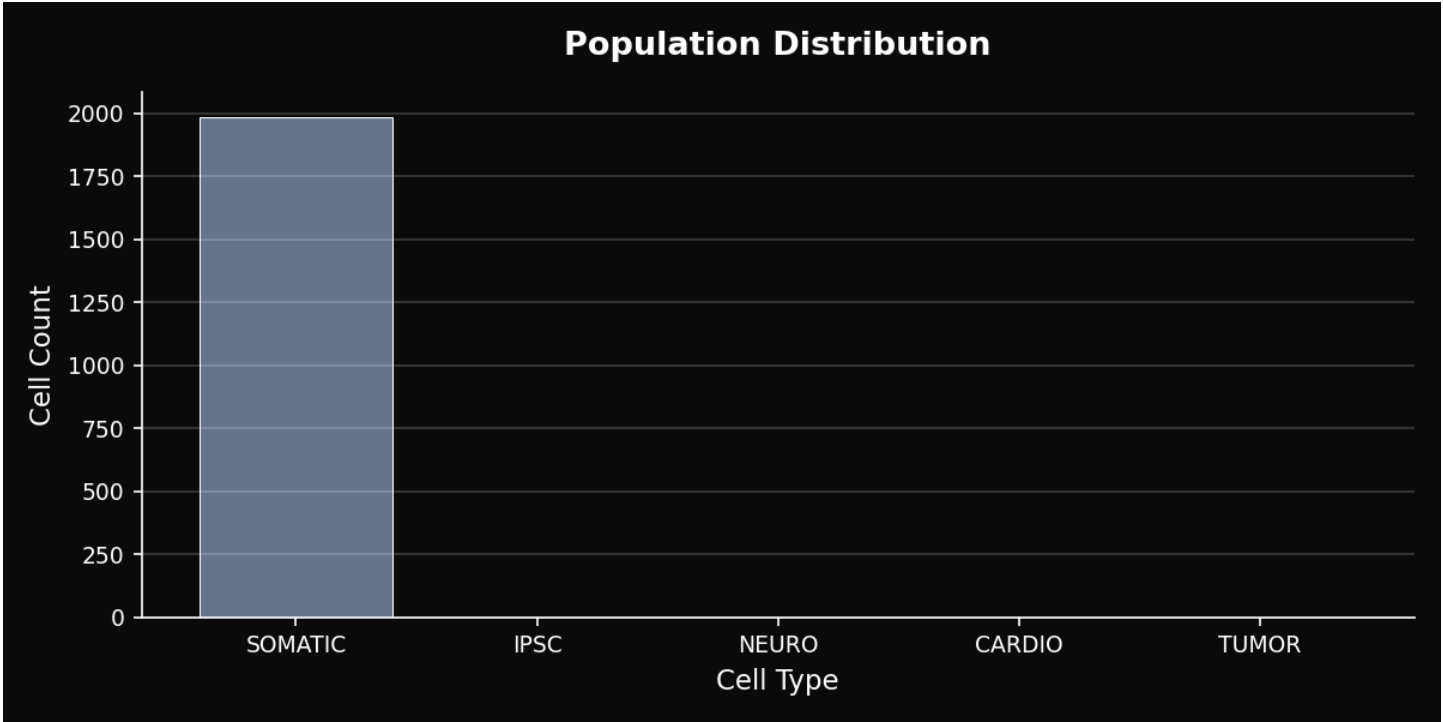
3. Results

The final cellular population consisted exclusively of 1987 somatic cells, with no induced pluripotent stem cells (iPSCs), neurogenic, cardiogenic, or tumorigenic cells observed. Despite the application of multiple rounds of OSKM transfection and direct cardiac transdifferentiation, the experimental conditions did not yield any detectable changes in cell type differentiation or reprogramming.

3.1 Population Distribution

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Cell Type	Count	Percentage
SOMATIC	1987	100.0%
IPSC	0	0.0%
NEURO	0	0.0%
CARDIO	0	0.0%
TUMOR	0	0.0%

3.2 Transcriptomic Analysis (scVI)

High-Resolution Gene Expression Analyses Performed: 2

Cells were expanded to high-dimensional transcriptomic space using a pre-trained variational autoencoder model on Human Cell Atlas data.

4. Discussion

The results indicate that the interventions employed were insufficient to drive significant transdifferentiation or reprogramming under the given conditions. The persistence of the somatic cell population suggests potential limitations in the reprogramming efficiency of the OSKM factors or cardiac transdifferentiation protocols used. These findings underscore the need for further optimization of reprogramming strategies, possibly by adjusting the timing, dosage, or combination of factors to enhance cellular plasticity and conversion efficiency.

This report was automatically generated by the IS-CHRP v25.0 Clinical AI Bridge platform. The data represents computational simulation results from a high-fidelity cellular reprogramming model.